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NEET · PHYSICS · CHAPTER-WISE PYQS

Semiconductor Electronics

NEET Physics · Previous Year Questions with Answer Key

119	37	2000 – 2026	23	English
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INSIDE THIS PDF

MCQ (Single Correct Answer)	96 questions
Out of syllabus	23 questions
Answer key	at the end

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1 **Solve**
Print it, or work through it on screen.

2 **Check**
The answer key is on the last pages.

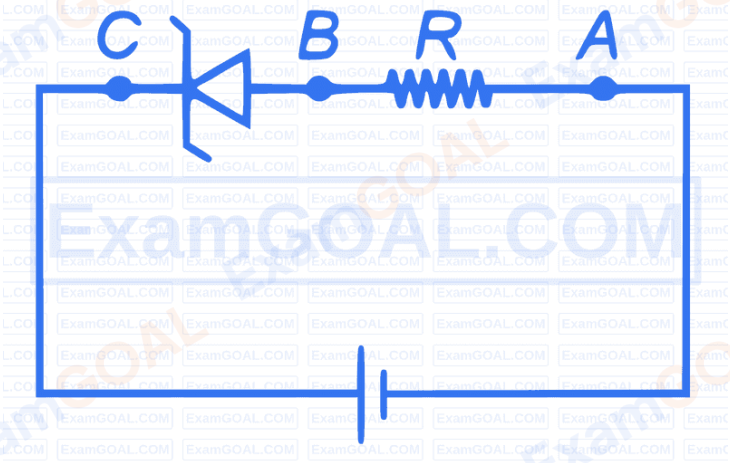
3 **Understand**
Scan the QR code beside any question for its full solution.

MCQ (Single Correct Answer) 96 questions

Q1. RE-NEET 2026

An ideal Zener diode with breakdown voltage of -3 V is reverse biased with a negative input voltage $V_i = -5$ V. The magnitude of voltage difference between point B and A is:

Open solution

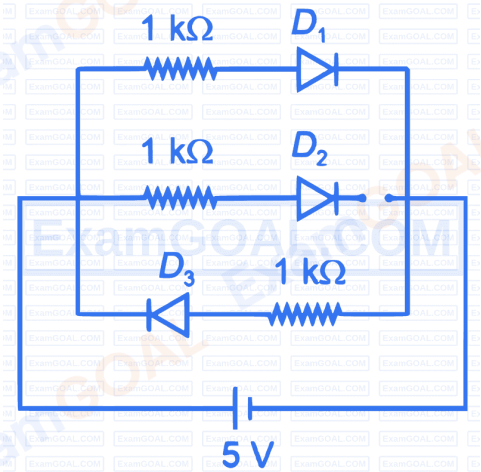


(A) 0 V (B) 3 V (C) 2 V (D) 1 V

Q2. RE-NEET 2026

Three identical p-n junction diodes D_1, D_2, and D_3 are connected across a battery as shown in the figure. If the width of the depletion regions of D_1, D_2 and D_3 are W_1, W_2 and W_3, respectively, then the correct option is:

Open solution



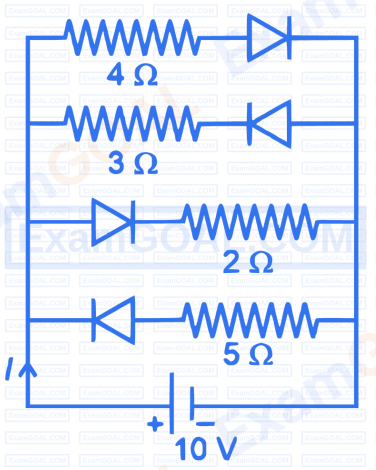
(A) $W_2 > W_1 = W_3$ (B) $W_1 > W_2 > W_3$ (C) $W_3 = W_1 > W_2$ (D) $W_3 > W_2 > W_1$

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Q3. NEET 2026

Open solution

The current / in the circuit shown below is: (All diodes are ideal and identical)



(A) $\frac{5}{3}$ A

(B) $\frac{5}{9}$ A

(C) $\frac{1}{3}$ A

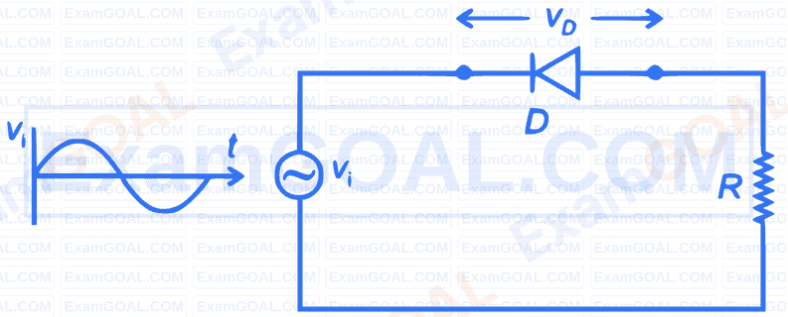
(D) $\frac{15}{2}$ A

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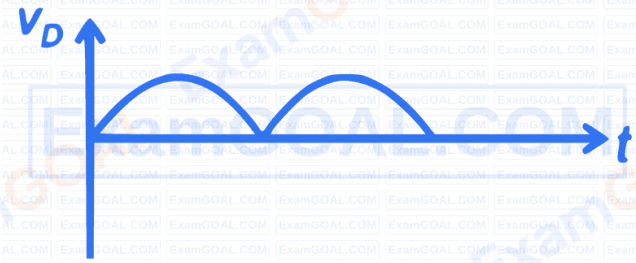
Q4. NEET 2026

Open solution

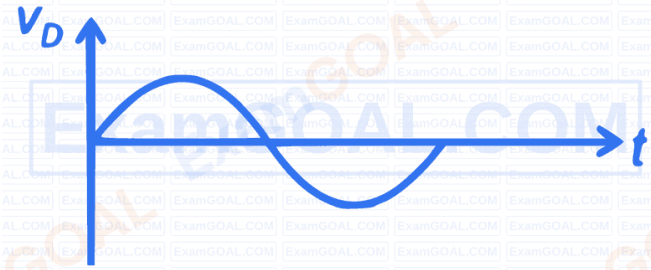
In the circuit shown below, the voltage appearing across the diode D will be of the form:



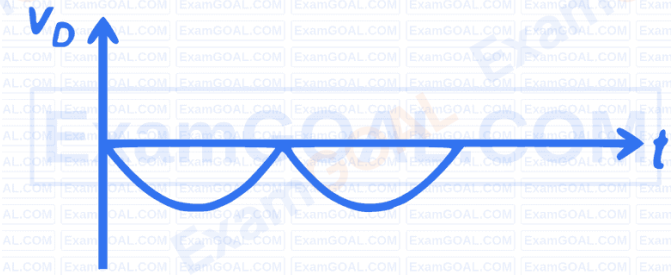
(A)



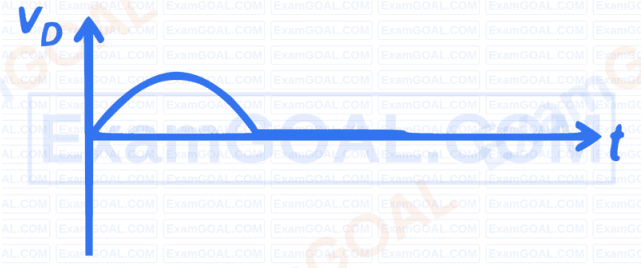
(B)



(C)



(D)



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Q5. NEET 2026

Open solution

Two statements are given below:

A. When the forward bias voltage across a p-n junction diode increases above a certain threshold voltage, the diode current increases significantly.

B. This current is called reverse saturation current.

Choose the correct answer from the options given below:

(A) Both Statements A and B are true

(B) Statement A is true, but Statement B is false

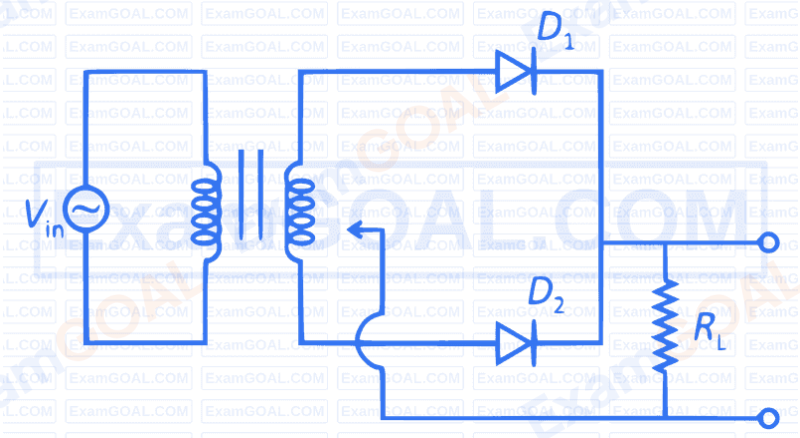
(C) Both Statements A and B are false

(D) Statement A is false, but Statement B is true

Q6. NEET 2025

Open solution

A full wave rectifier circuit with diodes (D_1) and (D_2) is shown in the figure. If input supply voltage $V_{in} = 220\sin(100\pi t)$ volt, then at $t = 15\text{msec}$



(A) D_1 and D_2 both are forward biased

(B) D_1 and D_2 both are reverse biased

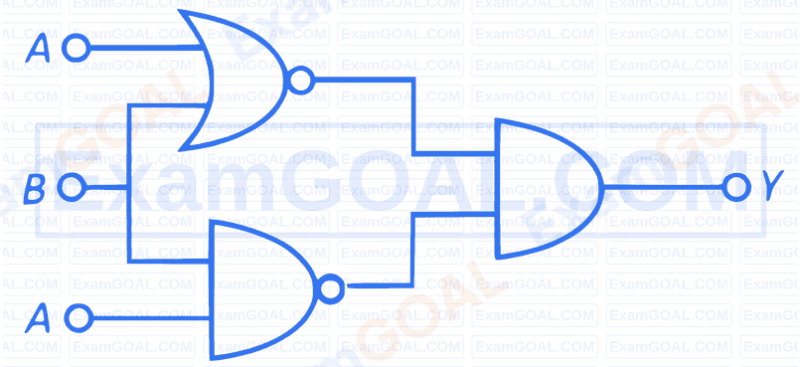
(C) D_1 is forward biased, D_2 is reverse biased

(D) D_1 is reverse biased, D_2 is forward biased.

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Q7. NEET 2025

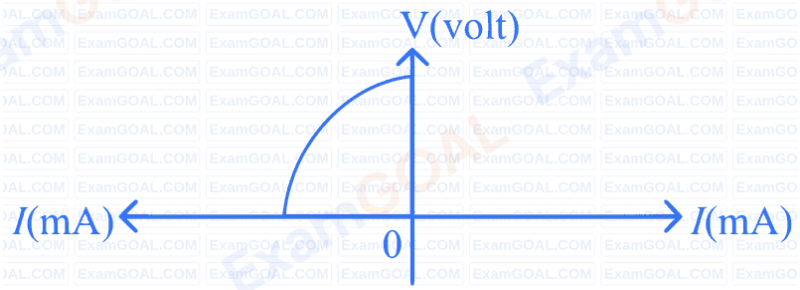
The output (Y) of the given logic implementation is similar to the output of an/a _____ gate.



Open solution

(A) OR (B) NOR (C) AND (D) NAND

Q8. NEET 2024 (Re-Examination)



Open solution

The I-V characteristics shown above are exhibited by a

(A) Light emitting diode (B) Zener diode
(C) Photodiode (D) Solar cell

Q9. NEET 2024 (Re-Examination)

When the output of an OR gate is applied as input to a NOT gate, then the combination acts as a

(A) NAND gate (B) NOR gate (C) AND gate (D) OR gate

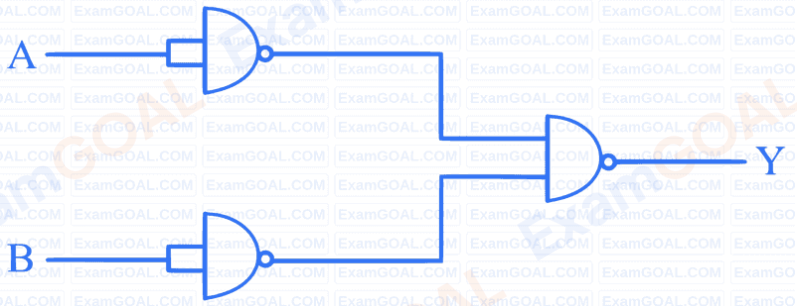
Open solution

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Q10. NEET 2024 (Re-Examination)

Open solution

The output Y for the inputs A and B of the given logic circuit is:



(A) $A \cdot B$ (B) $\bar{A} \cdot \bar{B}$ (C) $A + B$ (D) $\bar{A} + \bar{B}$

Q11. NEET 2024

Open solution

A logic circuit provides the output Y as per the following truth table :

A	B	Y
0	0	1
0	1	0
1	0	1
1	1	0

The expression of the output Y is :

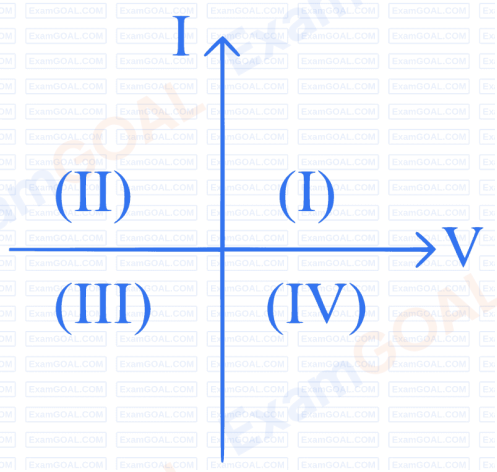
(A) $A \cdot B + \bar{A}$ (B) $A \cdot \bar{B} + \bar{A}$ (C) $\bar{B}$ (D) B

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Q12. NEET 2024

Open solution

Consider the following statements A and B and identify the correct answer :



A. For a solar-cell, the I-V characteristics lies in the IV quadrant of the given graph.

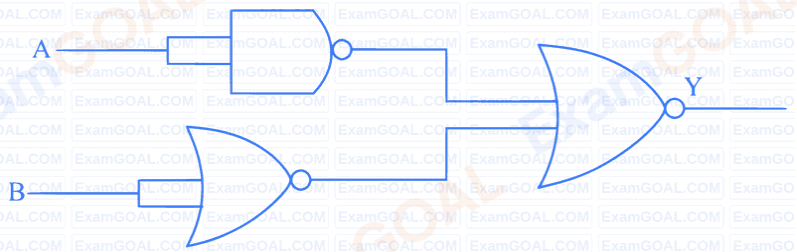
B. In a reverse biased pn junction diode, the current measured in (μA), is due to majority charge carriers.

(A) A is correct but B is incorrect
(B) A is incorrect but B is correct
(C) Both A and B are correct
(D) Both A and B are incorrect

Q13. NEET 2024

Open solution

The output (Y) of the given logic gate is similar to the output of an/a



(A) NAND gate
(B) NOR gate
(C) OR gate
(D) AND gate

Q14. NEET 2023 Manipur

Open solution

On the basis of electrical conductivity, which one of the following materials has the smallest resistivity?

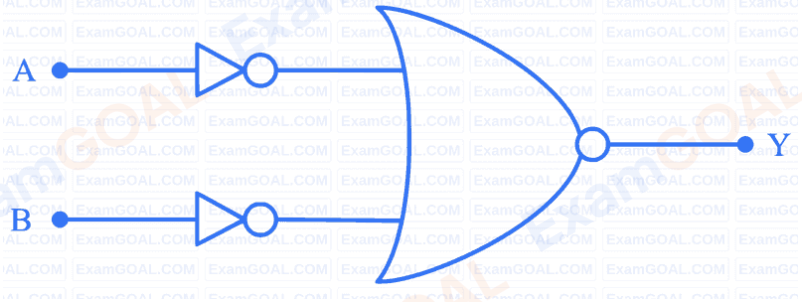
(A) Germanium
(B) Silver
(C) Glass
(D) Silicon

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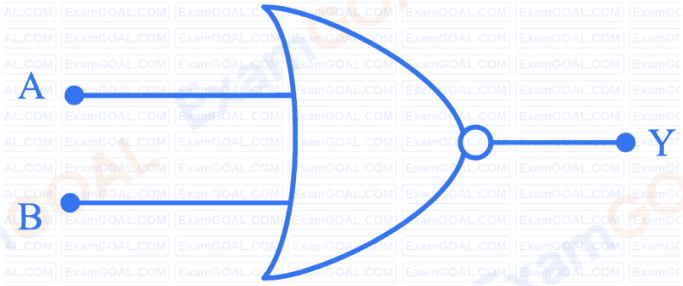
Q15. NEET 2023 Manipur

Open solution

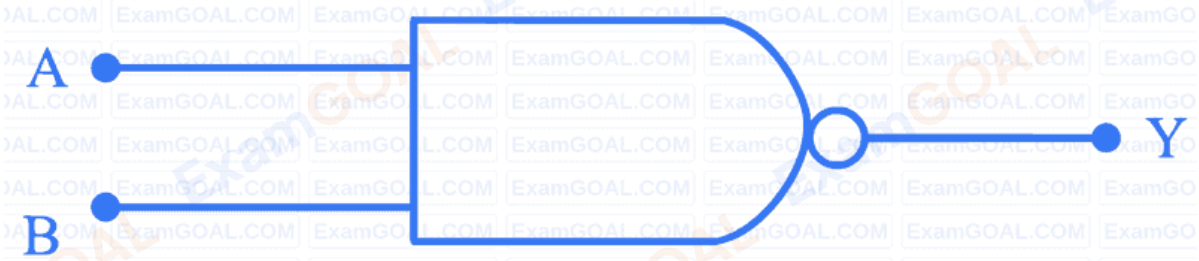
The given circuit is equivalent to:



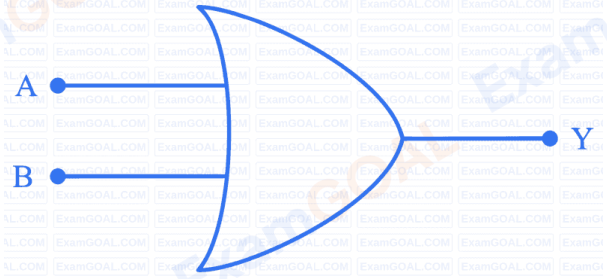
(A)



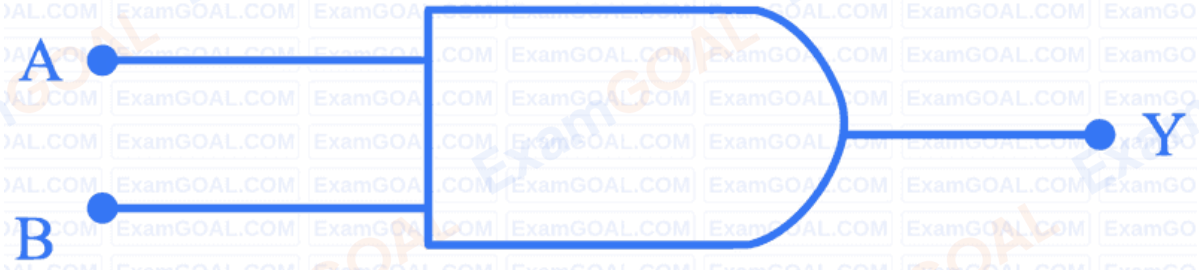
(B)



(C)



(D)



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Q16. NEET 2023 Manipur

A p-type extrinsic semiconductor is obtained when Germanium is doped with:

(A) Antimony (B) Phosphorous (C) Arsenic (D) Boron

Open solution

Q17. NEET 2023

Given below are two statements:

Statement I : Photovoltaic devices can convert optical radiation into electricity.

Statement II : Zener diode is designed to operate under reverse bias in breakdown region.

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) Both Statement I and Statement II are incorrect.

(B) Statement I is correct but Statement II is incorrect.

(C) Statement I is incorrect but Statement II is correct.

(D) Both Statement I and Statement II are correct

Open solution

Q18. NEET 2023

A full wave rectifier circuit consists of two p-n junction diodes, a centre-tapped transformer, capacitor and a load resistance. Which of these components remove the ac ripple from the rectified output?

(A) p-n junction diodes (B) Capacitor

(C) Load resistance (D) A centre-tapped transformer

Open solution

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Q19. NEET 2023

Open solution

For the following logic circuit, the truth table is:



(A)

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

(B)

A	B	Y
0	0	1
0	1	0
1	0	1
1	1	0

(C)

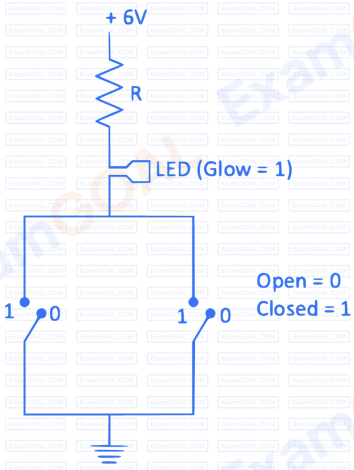
A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

(D)

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

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Q20. NEET 2022 Phase 2



Open solution

Identify the equivalent logic gate represented by the given circuit

(A) NAND (B) OR (C) NOR (D) AND

Q21. NEET 2022 Phase 2

The incorrect statement about the property of a Zener diode is :

(A) p and n regions of Zener diode are heavily doped

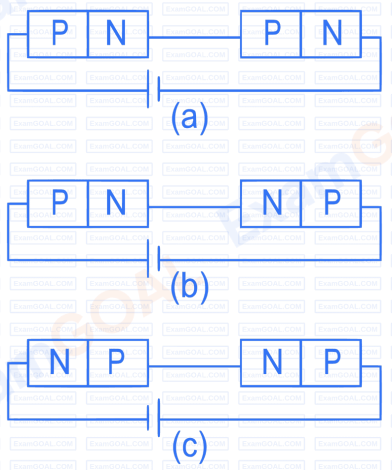
(B) Zener voltage remains constant at breakdown

(C) It is designed to operate under reverse bias

(D) Depletion region formed is very wide

Open solution

Q22. NEET 2022 Phase 1



Open solution

In the given circuits (a), (b) and (c), the potential drop across the two p-n junctions are equal in

(A) Circuit (a) only (B) Circuit (b) only

(C) Circuit (c) only (D) Both circuits (a) and (c)

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Q23. NEET 2022 Phase 1

As the temperature increases, the electrical resistance

(A) Increases for both conductors and semiconductors

(B) Decreases for both conductors and semiconductors

(C) Increases for conductors but decreases for semiconductors

(D) Decreases for conductors but increases for semiconductors

Open solution

Q24. NEET 2022 Phase 1

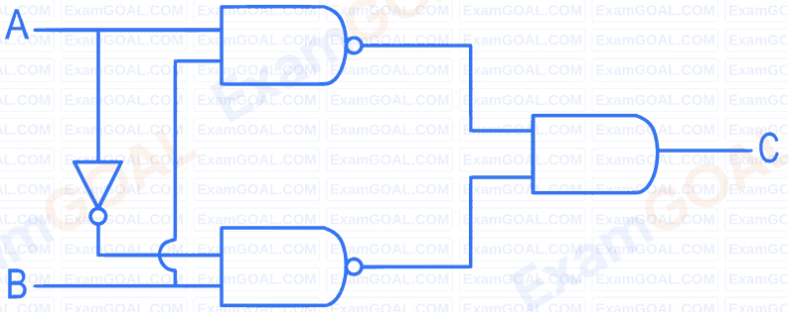
In half wave rectification, if the input frequency is 60 Hz, then the output frequency would be

(A) Zero (B) 30 Hz (C) 60 Hz (D) 120 Hz

Open solution

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Q25. NEET 2022 Phase 1



Open solution

The truth table for the given logic circuit is

(A)

A	B	C
0	0	0
0	1	1
1	0	1
1	1	0

(B)

A	B	C
0	0	1
0	1	0
1	0	0
1	1	1

(C)

A	B	C
0	0	1
0	1	0
1	0	1
1	1	0

(D)

A	B	C
0	0	0
0	1	1
1	0	0
1	1	1

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Q26. NEET 2021

Consider the following statements (A) and (B) and identify the correct answer.

(A) A Zener diode is connected in reverse bias, when used as a voltage regulator.

(B) The potential barrier of p-n junction lies between 0.1 V to 0.3 V.

Open solution

(A) (A) is incorrect but (B) is correct.
(B) (A) and (B) both are correct.
(C) (A) and (B) both are incorrect.
(D) (A) is correct and (B) is incorrect.

Q27. NEET 2021

The electron concentration in an n-type semiconductor is the same as hole concentration in a p-type semiconductor. An external field (electric) is applied across each of them. Compare the currents in them.

Open solution

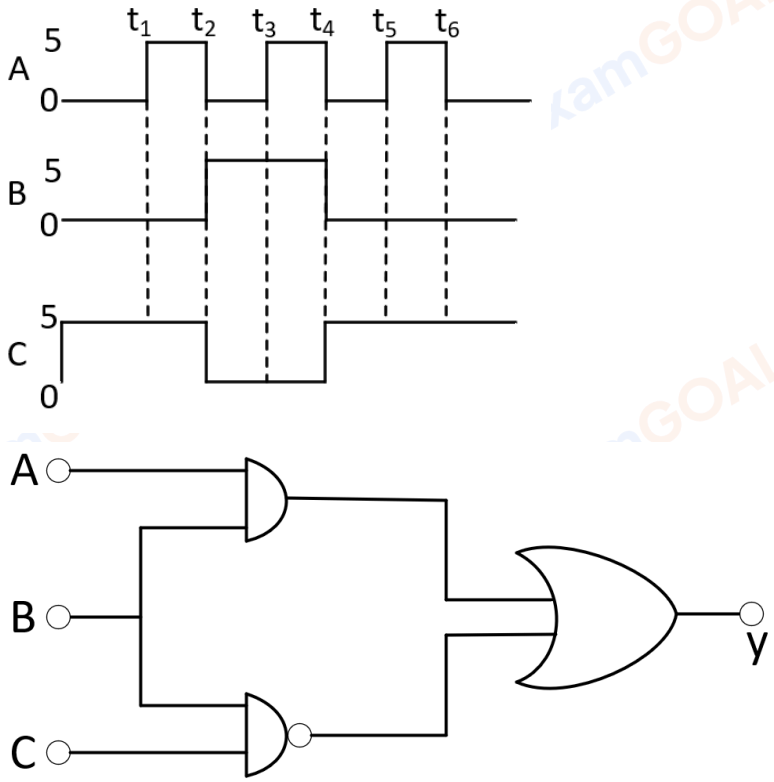
(A) No current will flow in p-type, current will only flow in n-type
(B) Current in n-type = current in p-type
(C) current in p-type > current in n-type
(D) current in n-type > current in p-type

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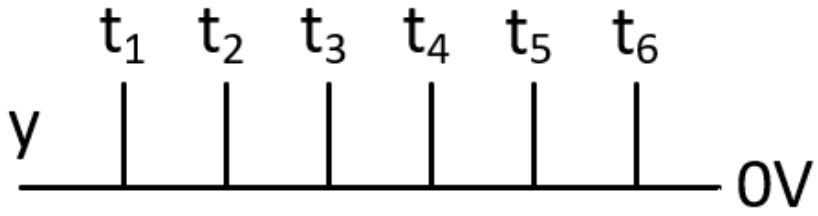
Q28. NEET 2021

For the given circuit, the input digital signals are applied at the terminals A, B and C. What would be the output at the terminal y?

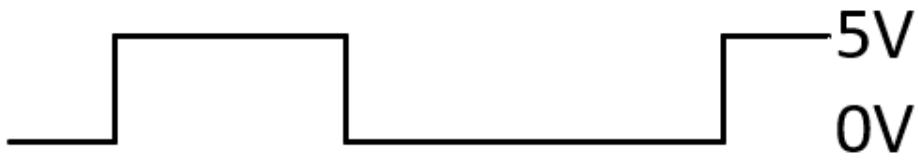
Open solution



(A)



(B)

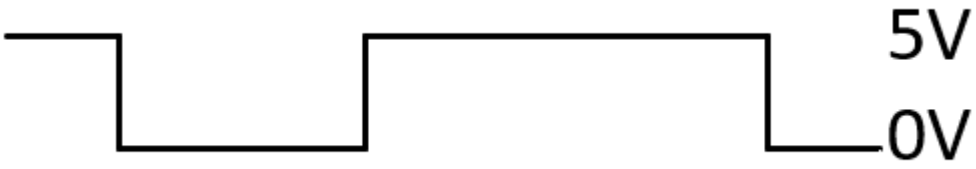


(C)



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(D)



Q29. NEET 2020 Phase 1

The increase in the width of the depletion region in a p-n junction diode is due to :

(A) reverse bias only

(B) both forward bias and reverse bias

(C) increase in forward current

(D) forward bias only

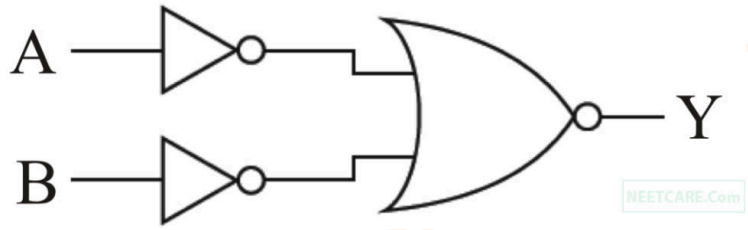
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Q30. NEET 2020 Phase 1

For the logic circuit shown, the truth table is :

Open solution



(A)

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

(B)

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

(C)

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

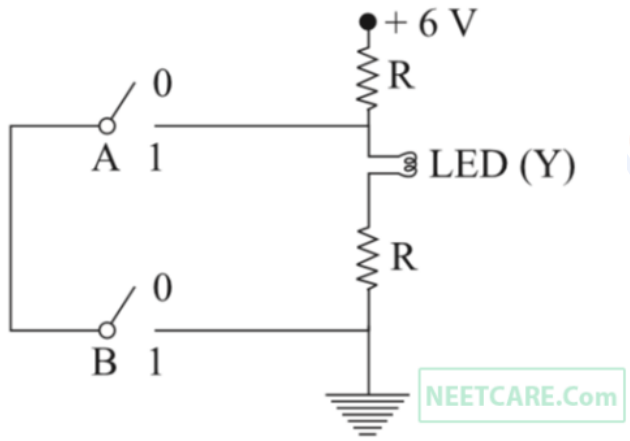
(D)

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

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Q31. NEET 2019

The **correct** Boolean operation represented by the circuit diagram drawn is :



Open solution

(A) OR (B) NAND (C) NOR (D) AND

Q32. NEET 2019

For a p-type semiconductor, which of the following statements is **true**?

(A) Holes are the majority carriers and trivalent atoms are the dopants.

(B) Holes are the majority carriers and pentavalent atoms are the dopants.

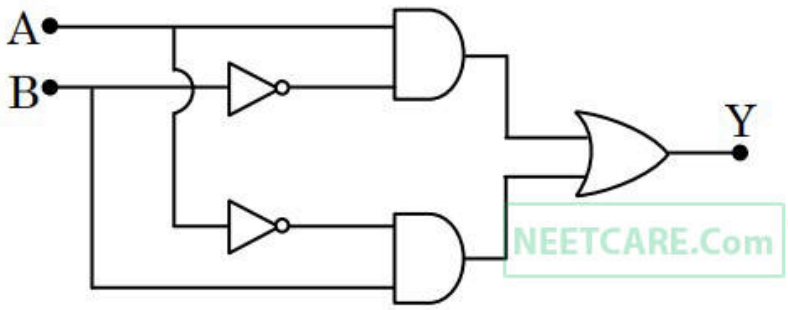
(C) Electrons are the majority carriers and pentavalent atoms are the dopants

(D) Electrons are the majority carriers and trivalent atoms are the dopants.

Open solution

Q33. NEET 2018

In the combination of the following gates the output Y can be written in terms of inputs A and B as



(A) $\overline{A.B}$ (B) $A.\overline{B}+\overline{A}.B$ (C) $\overline{A.B}+A.B$ (D) $\overline{A+B}$

Open solution

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Q34. NEET 2018

In a p-n junction diode, change in temperature due to heating

(A) affects only reverse resistance

(B) affects only forward resistance

(C) does not affect resistance of p-n junction

(D) affects the overall V - I characteristics of p-n junction

Open solution

Q35. NEET 2017

The given electrical network is equivalent to



(A) OR gate (B) NOR gate (C) NOT gate (D) AND gate

Open solution

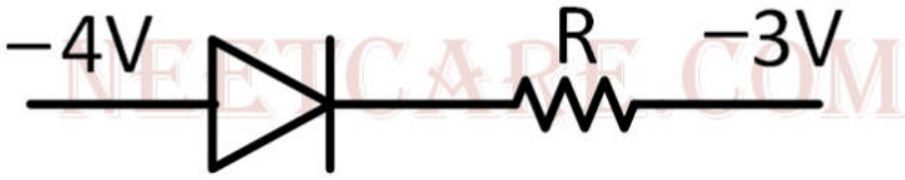
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Q36. NEET 2017

Open solution

Which one of the following represents forward bias diode ?

(A)



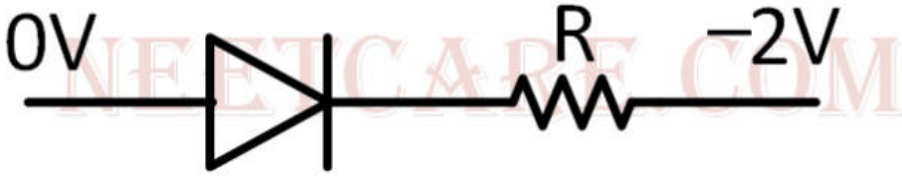
(B)



(C)



(D)

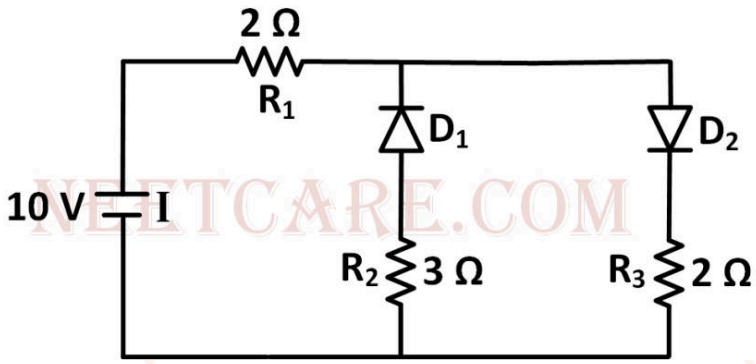


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Q37. NEET 2016 Phase 2

The given circuit has two ideal diodes connected as shown in the figure. The current flowing through the resistance R_1 will be

Open solution



(A) 2.5 A (B) 10.0 A (C) 1.43 A (D) 3.13 A

Q38. NEET 2016 Phase 2

What is the output Y in the following circuit, when all the three inputs A, B, C are first 0 and then 1 ?

Open solution

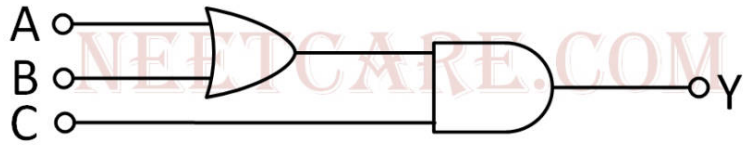


(A) 0, 1 (B) 0, 0 (C) 1, 0 (D) 1, 1

Q39. NEET 2016 Phase 1

To get output 1 for the following circuit, the correct choice for the input is

Open solution



(A) A = 1, B = 1, C = 0
(B) A = 1, B = 0, C = 1
(C) A = 0, B = 1, C = 0
(D) A = 1, B = 0, C = 0

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Q40. NEET 2016 Phase 1

Consider the junction diode as ideal. The value of current flowing through AB is

Open solution

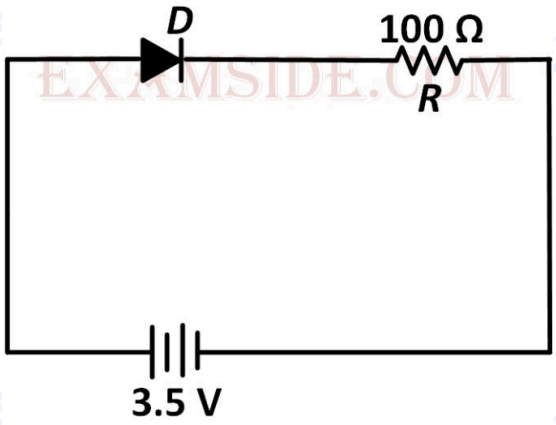


(A) 10^{-1} A (B) 10^{-3} A (C) 0 A (D) 10^{-2} A

Q41. AIPMT 2015

In the given figure, a diode D is connected to an external resistance R = 100 Ω and an e.m.f. of 3.5 V. If the barrier potential developed across the diode is 0.5 V, the current in the circuit will be

Open solution



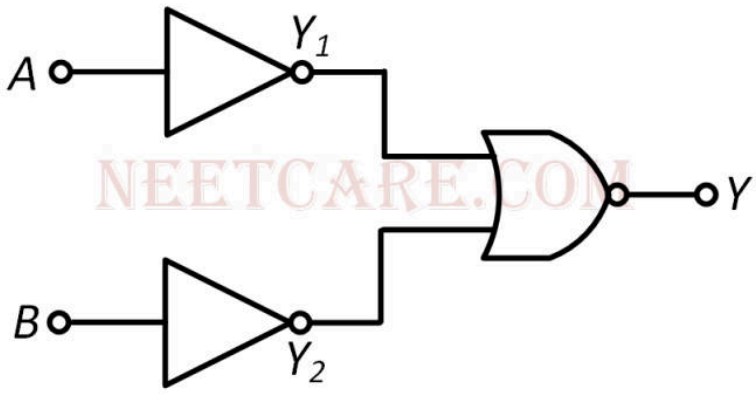
(A) 20 mA (B) 35 mA (C) 30 mA (D) 40 mA

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Q42. AIPMT 2015 Cancelled Paper

Open solution

Which logic gate is represented by the following combination of logic gates ?



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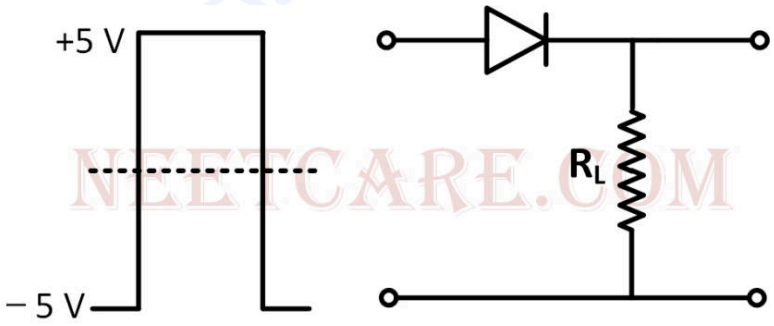
(A) AND (B) NOR (C) OR (D) NAND

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Q43. AIPMT 2015 Cancelled Paper

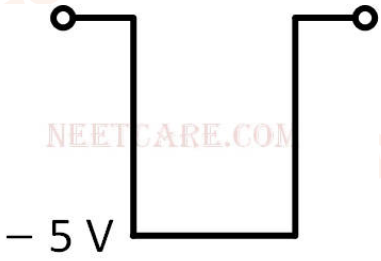
Open solution

If in a p-n junction, a square input signal of 10 V is applied, as shown,

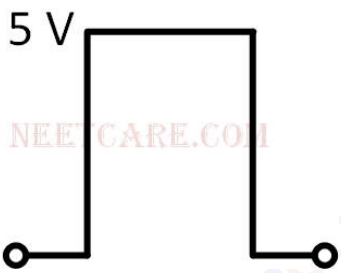


then the output across R_L will be

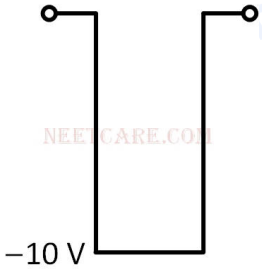
(A)



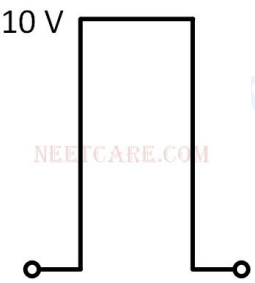
(B)



(C)



(D)

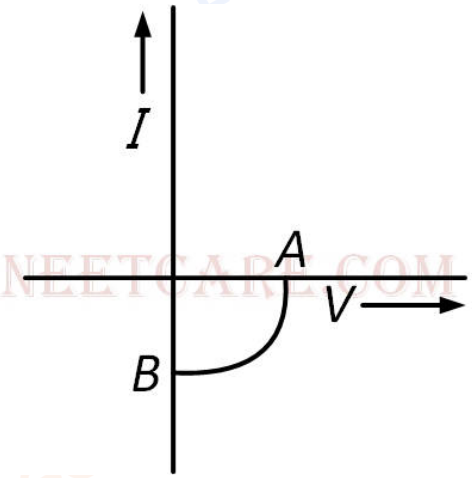


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Q44. AIPMT 2014

Open solution

The given graph represents V-I characteristic for a semiconductor device.



Which of the following statement is correct ?

(A) It is V-I characteristic for solar cell where, point A represents open circuit voltage and point B short circuit current.

(B) It is for a solar cell and popints A and B represent open circuit voltage and current, respectively.

(C) It is for a photodiode and points A and B represent open circuit voltage and current, respectively.

(D) It is for a LED and points A and B represent open circuit voltage and short circuit current, respectively.

Q45. AIPMT 2014

Open solution

The barrier potential of a p-n junction depends on

(1) type of semiconductor material

(2) amount of doping

(3) temperature

Which one of the following is correct ?

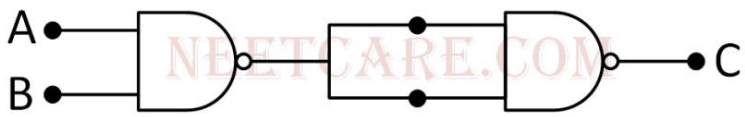
(A) (1) and (2) only **(B)** (2) only **(C)** (2) and (3) only **(D)** (1), (2) and (3)

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Q46. NEET 2013 (Karnataka)

The output from a NAND gate is divided into two in parallel and fed to another NAND gate. The resulting gate is a

Open solution



(A) AND gate (B) NOR gate (C) OR gate (D) NOT gate

Q47. NEET 2013 (Karnataka)

In an unbiased p-n junction, holes diffuse from the p-region to n-region because of

(A) The attraction of free electrons of n-region.

(B) The higher hole concentration in p-region than that in n-region.

(C) The higher concentration of electrons in the n-region than that in the p-region.

(D) The potential difference across the p-n junction.

Open solution

Q48. NEET 2013

In a n-type semiconductor, which of the following statements is true?

(A) Holes are minority carriers and pentavalent atoms are dopants.

(B) Holes are majority carriers and trivalent atoms are dopants.

(C) Electrons are majority carriers and trivalent atoms are dopants.

(D) Electrons are minority carriers and pentavalent atoms are dopants.

Open solution

Q49. NEET 2013

The output (X) of the logic circuit shown in figure will be

Open solution



(A) X = A . B

(B) X = $\overline{A+B}$

(C) X = $\overline{\overline{A}.\overline{B}}$

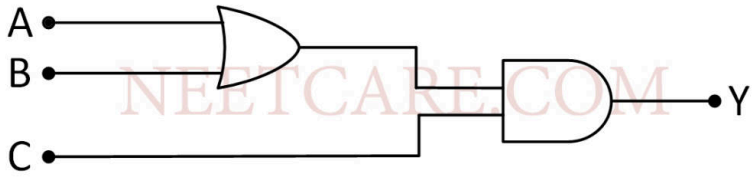
(D) X = $\overline{A.B}$

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Q50. AIPMT 2012 Mains

To get an output Y= 1 in given circuit which of the following input will be correct ?

Open solution



(A) A → 1, B → 0, C → 0

(B) A → 1, B → 0, C → 1

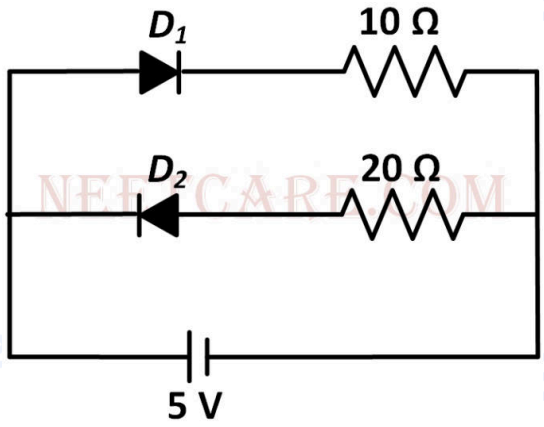
(C) A → 1, B → 1, C → 0

(D) A → 0, B → 1, C → 0

Q51. AIPMT 2012 Prelims

Two ideal diodes are connected to a battery as shown in the circuit. The current supplied by the battery is

Open solution



(A) 0.75 A

(B) zero

(C) 0.25 A

(D) 0.5 A

Q52. AIPMT 2012 Prelims

C and Si both have same lattice structure; having 4 bonding electrons in each. However, C is insulator where as Si is intrinsic semiconductor. This is because

Open solution

(A) In case of C the valence band is not completely filled at absolute zero temperature.

(B) In case of C the conduction band is partly filled even at absolute zero temperature.

(C) The four bonding electrons in the case of C lie in the second orbit, whereas in the case of Si they lie in the third.

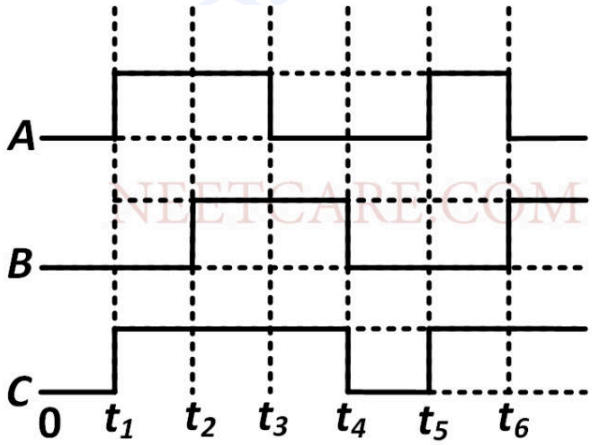
(D) The four bonding electrons in the case of C lie in the third orbit , whereas for Si they lie in the fourth orbit.

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Q53. AIPMT 2012 Prelims

Open solution

The figure shows a logic circuit with two inputs A and B and the output C. The voltage wave forms across A, B and C are as given. The logic circuit gate is

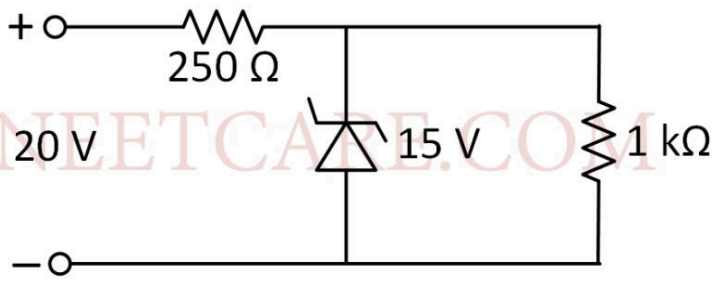


(A) OR gate (B) NOR gate (C) AND gate (D) NAND gate

Q54. AIPMT 2011 Mains

Open solution

A Zener diode, having breakdown voltage equal to 15 V, is used in a voltage regulator circuit shown in figure. The current through the diode is

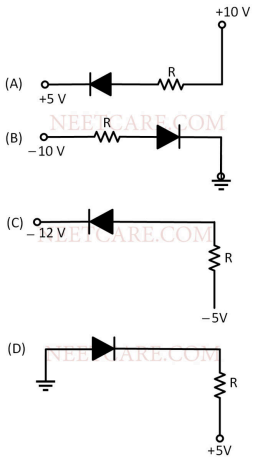


(A) 5 mA (B) 10 mA (C) 15 mA (D) 20 mA

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Q55. AIPMT 2011 Mains

In the following figure, the diodes which are forward biased, are



Open solution

(A) (A), (B) and (D) **(B)** (C) only **(C)** (C) and (A) **(D)** (B) and (D)

Q56. AIPMT 2011 Mains

Pure Si at 500 K has equal number of electron (n_e) and hole (n_h) concentrations of 1.5×10^{16} m^{-3}. Doping by indium increases n_h to 4.5×10^{22} m^{-3}. The doped semiconductor is of

Open solution

(A) p-type having electron concentration $n_e = 5 \times 10^9$ m^{-3}

(B) n-type with electron concentration $n_e = 5 \times 10^{22}$ m^{-3}

(C) p-type with electron concentration $n_e = 2.5 \times 10^{10}$ m^{-3}

(D) n-type with electron concentration $n_e = 2.5 \times 10^{23}$ m^{-3}

Q57. AIPMT 2011 Prelims

In forward biasing of the p-n junction

Open solution

(A) the positive terminal of the battery is connected to p-side and the depletion region becomes thick.

(B) the positive terminal of the battery is connected to n-side and the depletion region becomes thin.

(C) the positive terminal of the battery is connected to n-side and the depletion region becomes thick.

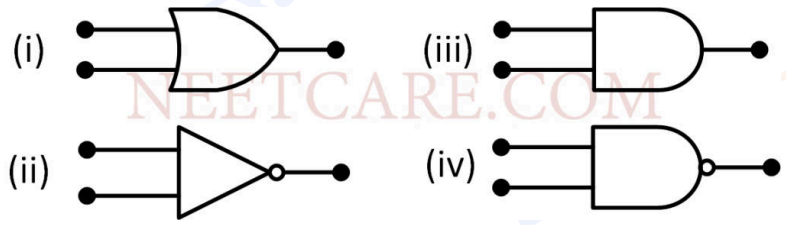
(D) the positive terminal of the battery is connected to p-side and the depletion region becomes thin.

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Q58. AIPMT 2011 Prelims

Open solution

Symbolic representation of four logic gates are shown as



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Pick out which ones are for AND, NAND and NOT gates, respectively

(A) (ii), (iii) and (iv)
(B) (iii), (ii) and (i)
(C) (iii), (ii) and (iv)
(D) (iii), (iv) and (ii)

Q59. AIPMT 2011 Prelims

Open solution

If a small amount of antimony is added to germanium crystal

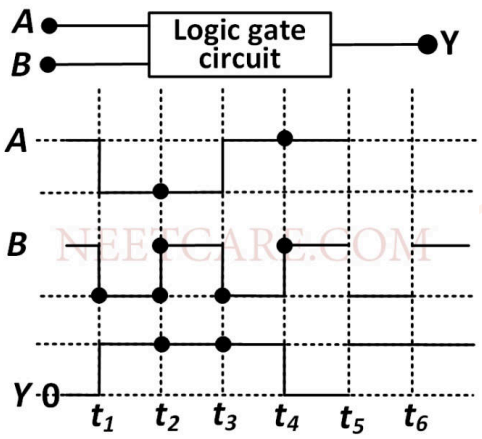
(A) it becomes a p-type semiconductor
(B) the antimony becomes an acceptor atom
(C) there will be more free electrons than holes in the semiconductor
(D) its resistance is increased

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Q60. AIPMT 2010 Mains

The following figure shows a logic gate circuit with two inputs A and B and the output Y. The voltage waveforms of A, B and Y are as given.

Open solution



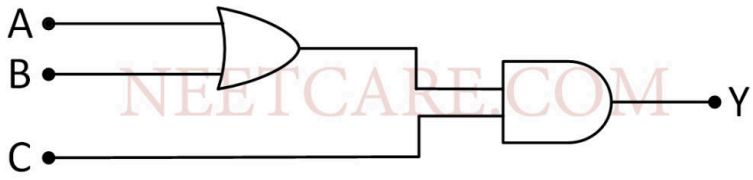
The logic gate is

(A) NOR gate (B) OR gate (C) AND gate (D) NAND gate

Q61. AIPMT 2010 Prelims

To get an output Y = 1 in given circuit which of the following input will be correct ?

Open solution



(A) A $\rightarrow$ 1, B $\rightarrow$ 0, C $\rightarrow$ 0

(B) A $\rightarrow$ 1, B $\rightarrow$ 0, C $\rightarrow$ 1

(C) A $\rightarrow$ 1, B $\rightarrow$ 1, C $\rightarrow$ 0

(D) A $\rightarrow$ 0, B $\rightarrow$ 1, C $\rightarrow$ 0

Q62. AIPMT 2010 Prelims

Which one of the following statement is false ?

Open solution

(A) Pure Si doped with trivalent impurities gives a p-type semiconductor.

(B) Majority carries in a n-type semiconductor are holes.

(C) Minority carries in a p-type semiconductor are electrons.

(D) The resistance of intrinisic semiconductor decreases with increase of temperature.

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Q63. AIPMT 2010 Prelims

Which one of the following bonds produces a solid that reflects light in the visible region and whose electrical conductivity decreases with temperature and has high melting point ?

Open solution

(A) metallic bonding
(B) van der Waal's bonding
(C) ionic bonding
(D) covalent bonding

Q64. AIPMT 2010 Prelims

The device that can act as a complete electronic circuit is

Open solution

(A) junction diode
(B) integrated circuit
(C) junction transistor
(D) zener diode

Q65. AIPMT 2009

A p-n photodiode is fabricated from a semiconductor with a band gap of 2.5 eV. It can detect a signal of wavelength

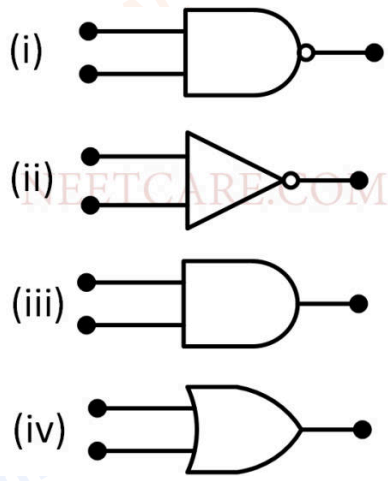
Open solution

(A) 4000 nm
(B) 6000 nm
(C) 4000 $\mathring{A}$
(D) 6000 $\mathring{A}$

Q66. AIPMT 2009

The symbolic representation of four logic gates are given below

Open solution



The logic symbols for OR, NOT and NAND gates are respectively

(A) (iv), (i), (iii)
(B) (iv), (ii), (i)
(C) (i), (iii), (iv)
(D) (iii), (iv), (ii)

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Q67. AIPMT 2009

Sodium has body centred packing. Distance between two nearest atoms is 3.7 $\mathring{A}$. The lattice parameter is

(A) 4.3 $\mathring{A}$
(B) 3.0 $\mathring{A}$
(C) 8.6 $\mathring{A}$
(D) 6.8 $\mathring{A}$

Open solution

Q68. AIPMT 2008

A p-n photodiode is made of a material with a band gap of 2.0 eV. The minimum frequency of the radiation that can be absorbed by the material is nearly

(A) 1×10^{14} Hz
(B) 20×10^{14} Hz
(C) 10×10^{14} Hz
(D) 5×10^{14} Hz

Open solution

Q69. AIPMT 2008

If the lattice parameter for a crystalline structure is 3.6 $\mathring{A}$, then the atomic radius in fcc crystal is

(A) 2.92 $\mathring{A}$
(B) 1.27 $\mathring{A}$
(C) 1.81 $\mathring{A}$
(D) 2.10 $\mathring{A}$

Open solution

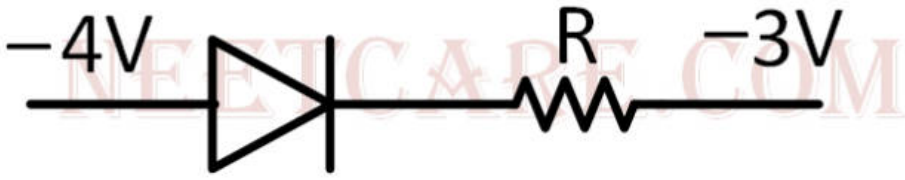
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Q70. AIPMT 2007

Open solution

Which one of the following represents forward bias diode ?

(A)



(B)



(C)



(D)

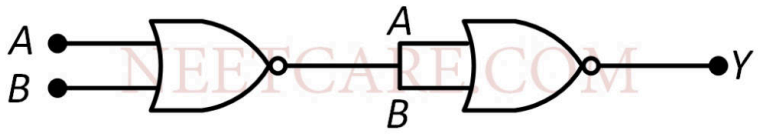


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Q71. AIPMT 2007

In the following circuit, the output Y for all possible inputs A and B is expressed by the truth table.

Open solution



(A)

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

(B)

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

(C)

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

(D)

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

Q72. AIPMT 2007

For a cubic crystal structure which one of the following relations indicating the cell characteristics is correct ?

Open solution

(A) a $\neq$ b $\neq$ c and $\alpha = \beta = \gamma = 90^o$

(B) a = b = c and $\alpha \neq \beta \neq \gamma = 90^o$

(C) a = b = c and $\alpha = \beta = \gamma = 90^o$

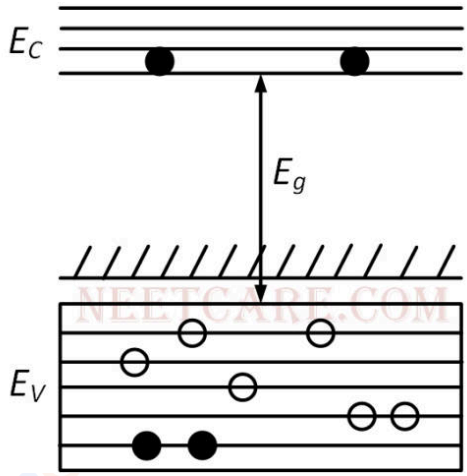
(D) a $\neq$ b $\neq$ c and ;$\alpha \neq \beta \neq$ and $\gamma = 90^o$

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Q73. AIPMT 2007

In the energy band diagram of a material shown below, the open circles and filled circles denote holes and electrons respectively. The material is

Open solution



(A) an insulator

(B) a metal

(C) an n-type semiconductor

(D) a p-type semiconductor.

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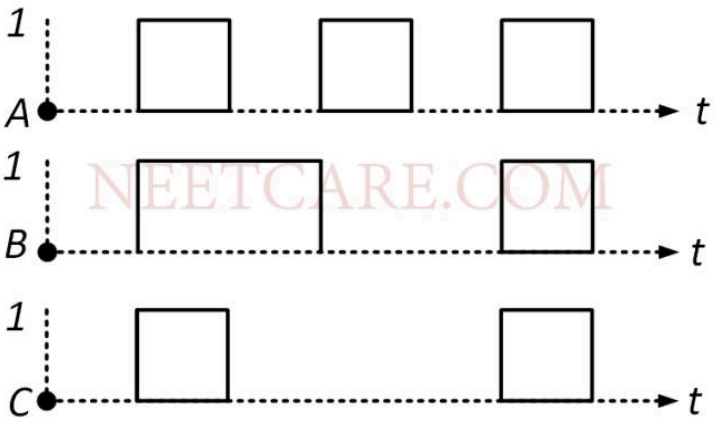
Q74. AIPMT 2006

Open solution

The following figure shows a logic gate circuit with two inputs A and B and the output C.



The voltage waveforms of A, B and C are as shown below.



The logic circuit gate is

(A) OR gate (B) AND gate (C) NAND gate (D) NOR gate

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Q75. AIPMT 2006

Which one of the following represents forward bias diode ?

Open solution

(A)



(B)



(C)



(D)



Q76. AIPMT 2005

Choose the only false statement from the following.

Open solution

(A) In conductors the valence and conduction bands overlap.

(B) Substances with energy gap of the order of 10 eV are insulators.

(C) The resistivity of a semiconductor increases with increase in temperature.

(D) The conductivity of a semiconductor increases with increase in temperature.

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Q77. AIPMT 2005

Zener diode is used for

(A) amplification

(B) rectification

(C) stabilisation

(D) producing oscillations in an oscillator.

Open solution

Q78. AIPMT 2005

Application of a forward bias to a p-n junction

(A) widens the depletion zone

(B) increases the potential difference across the depletion zone

(C) increases the number of donors on the n side

(D) decreases the electric field in the depletion zone.

Open solution

Q79. AIPMT 2005

Carbon, silicon and germanium atoms have four valence electrons each. Their valence and conduction bands are separated by energy band gaps represented by $(E_g)_C$, $(E_g)_{si}$ and $(E_g)_{Ge}$ respectively. Which one of the following relationships is true in their case?

(A) $(E_g)_C > (E_g)_{Si}$

(B) $(E_g)_C < (E_g)_{Si}$

(C) $(E_g)_C = (E_g)_{Si}$

(D) $(E_g)_C < (E_g)_{Ge}$.

Open solution

Q80. AIPMT 2005

Copper has face centered cubic (fcc) lattice with interatomic spacing equal to 2.54 $\mathring{A}$. The value of lattice constant for this lattice is

(A) 2.54 $\mathring{A}$

(B) 3.59 $\mathring{A}$

(C) 1.27 $\mathring{A}$

(D) 5.08 $\mathring{A}$

Open solution

Q81. AIPMT 2005

In a p-n junction photo cell, the value of the photo-electromotive force produced by monochromatic light is proportional to

(A) The barrier voltage at the p-n junction.

(B) The intensity of the light falling on the cell.

(C) The frequency of the light falling on the cell.

(D) The voltage applied at the p-n junction.

Open solution

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Q82. AIPMT 2004

In semiconductors at a room temperature

(A) the valence band is partially empty and the conduction band is partially filled

(B) the valence band is completely filled and the conduction band is partially filled

(C) the valence band is completely filled

(D) the conduction band is completely empty

Open solution

Q83. AIPMT 2004

The peak voltage in the output of a half wave diode rectifier fed with a sinusoidal signal without filter is 10 V. The d.c. component of the output voltage is

(A) $(10/\sqrt{2})$ V

(B) $\left(\frac{10}{\pi}\right)V$

(C) 10 V

(D) $(20/\pi)$ V

Open solution

Q84. AIPMT 2004

The output of OR gate is 1

(A) if both inputs are zero

(B) if either or both inputs are 1

(C) only if both inputs are 1

(D) if either inputs is zero

Open solution

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Q85. AIPMT 2004

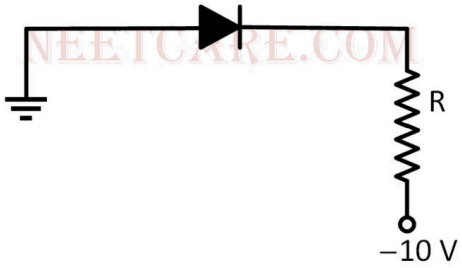
Open solution

Of the diodes shown in the following diagrams, which one is reverse biased ?

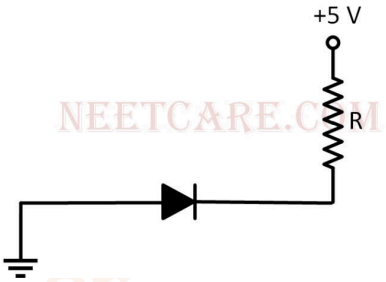
(A)



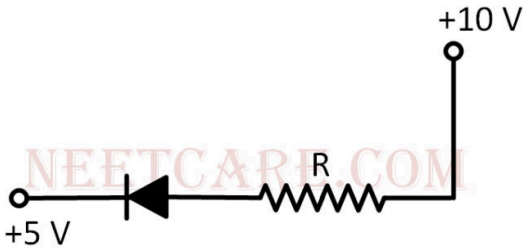
(B)



(C)



(D)



Q86. AIPMT 2003

Open solution

Reverse bias applied to a junction diode

(A) lowers the potential barrier

(B) raises the potential barrier

(C) increases the majority carrier current

(D) increases the minority carrier current

Q87. AIPMT 2003

Open solution

If a full wave rectifier circuit is operating from 50 Hz mains, the fundamental frequency in the ripple will be

(A) 25 Hz

(B) 50 Hz

(C) 70.7 Hz

(D) 100 Hz

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Q88. AIPMT 2003

Barrier potential of a p-n junction diode does not depened on

(A) diode design (B) temperature (C) forward bias (D) doping density

Open solution

Q89. AIPMT 2002

Number of atom per unit cell in B.C.C.

(A) 9 (B) 4 (C) 2 (D) 1

Open solution

Q90. AIPMT 2002

In a p-n junction

(A) high potential at n side and low potential at p side

(B) high potential at P side and low potential at n side

(C) p and n both are at same potential

(D) undetermined.

Open solution

Q91. AIPMT 2002

The given truth table is for which logic gate

A	B	Y
1	1	0
0	1	1
1	0	1
0	0	1

(A) NAND (B) XOR (C) NOR (D) OR.

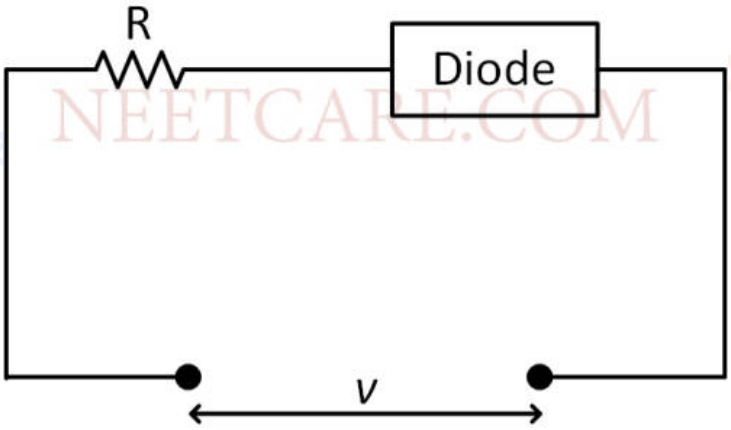
Open solution

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Q92. AIPMT 2002

For the given circuit of p-n junction diode which is correct

Open solution



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(A) in forward bias the voltage across R is V

(B) in reverse bias the voltage across R is V

(C) in forward bias the voltage across R is 2V

(D) in reverse bias the voltage across R is 2V.

Q93. AIPMT 2001

The given truth table is for which logic gate

Open solution

A	B	Y
1	1	0
0	1	1
1	0	1
0	0	1

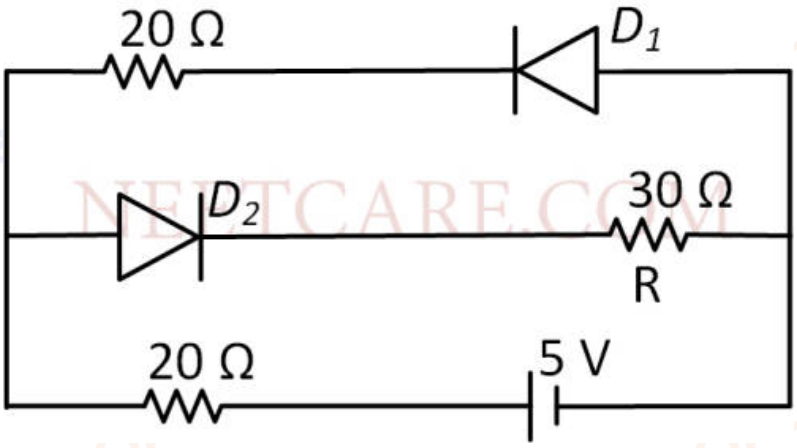
(A) NAND (B) XOR (C) NOR (D) OR

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Q94. AIPMT 2001

The current in the circuit will be

Open solution



(A) 5/40 A (B) 5/50 A (C) 5/10 A (D) 5/20 A

Q95. AIPMT 2000

The cations and anions are arranged in alternate form in

Open solution

(A) metallic crystal
(B) ionic crystal
(C) covalent crystal
(D) semi-conductor crystal.

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Q96. AIPMT 2000

From the following diode circuit, which diode is in forward biased condition

Open solution

(A)



(B)



(C)



(D)



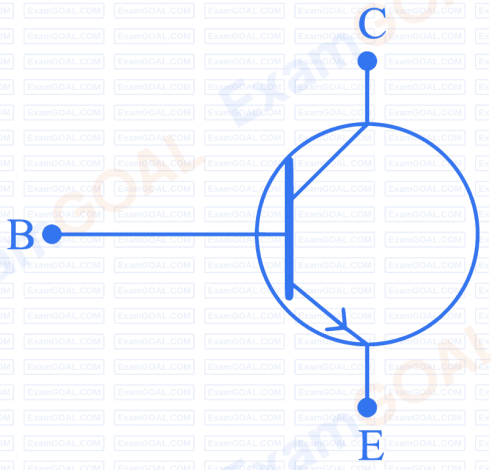
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Q97. NEET 2023 Manipur · MCQ (Single Correct Answer)

Open solution



The above figure shows the circuit symbol of a transistor. Select the correct statements given below:

(A) The transistor has two segments of p-type semiconductor separated by a segment of n-type semiconductor.

(B) The emitter is of moderate size and heavily doped.

(C) The central segment is thin and lightly doped.

(D) The emitter base junction is reverse biased in common emitter amplifier circuit.

(A) (C) and (D) **(B)** (A) and (D) **(C)** (A) and (B) **(D)** (B) and (C)

Q98. NEET 2022 Phase 2 · MCQ (Single Correct Answer)

Open solution

The collector current in a common base amplifier using n-p-n transistor is 24 mA. If 80% of the electrons released by the emitter is accepted by the collector, then the base current is numerically :

(A) 3 mA and entering the base **(B)** 6 mA and leaving the base

(C) 3 mA and leaving the base **(D)** 6 mA and entering the base

Q99. NEET 2020 Phase 1 · MCQ (Single Correct Answer)

Open solution

For transistor action, which of the following statements is **correct**?

(A) Base, emitter and collector regions should have same size.

(B) Both emitter junction as well as the collector junction are forwarded biased.

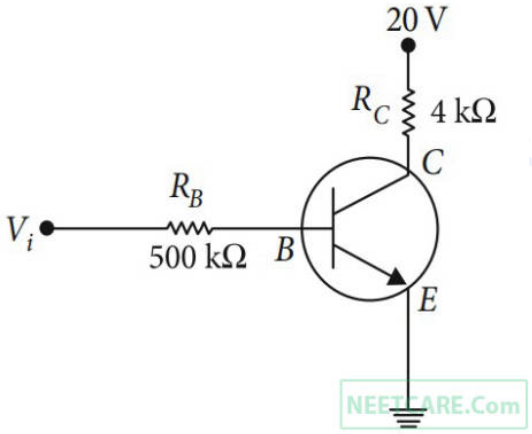
(C) The base region must be very thin and lightly doped.

(D) Base, emitter and collector regions should have same doping concentrations.

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Q100. NEET 2018 · MCQ (Single Correct Answer)

In the circuit shown in the figure, the input voltage V_i is 20 V, V_{BE} = 0 and V_{CE} = 0. The values of I_B, I_C and β are given by



Open solution

(A) I_B = 40 μA, I_C = 10 mA, β = 250

(B) I_B = 25 μA, I_C = 5 mA, β = 200

(C) I_B = 20 μA, I_C = 5 mA, β = 250

(D) I_B = 40 μA, I_C = 5 mA, β = 125

Q101. NEET 2017 · MCQ (Single Correct Answer)

In a common emitter transistor amplifier the audio sgnal voltage across the collector is 3 V. The resistance of collector is 3 kΩ. If current gain is 100 and the base resistance is 2 kΩ, the voltage and power gain of the amplifier is

Open solution

(A) 15 and 200 (B) 150 and 15000 (C) 20 and 2000 (D) 200 and 1000

Q102. NEET 2016 Phase 2 · MCQ (Single Correct Answer)

For CE transistor amplifier, the audio signal voltage across the collector resistance of 2 kΩ is 4 V. If the current amplification factor of the transistor is 100 and the base resistance is 1 kΩ, then the input signal voltage is

Open solution

(A) 10 mV (B) 20 mV (C) 30 mV (D) 15 mV

Q103. NEET 2016 Phase 1 · MCQ (Single Correct Answer)

A npn transistor is connected in common emitter configuration in a given amplifier. A load resistance of 800 Ω is connected in the collector circuit and the voltage drop across it is 0.8 V. If the current amplification factor is 0.96 and the input resistance of the circuit is 192 Ω, the voltage gain and the power gain of the amplifier will respectively be

Open solution

(A) 4, 4 (B) 4, 3.69 (C) 4, 3.84 (D) 3.69, 3.84

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Q104. AIPMT 2015 · MCQ (Single Correct Answer)

The input signal given to a CE amplifier having a voltage gain of 150 is $V_i = 2\cos(15t + \frac{\pi}{3})$. The corresponding output signal will be

(A) $2\cos\left(15t + \frac{5\pi}{6}\right)$
(B) $300\cos\left(15t + \frac{4\pi}{3}\right)$
(C) $300\cos\left(15t + \frac{\pi}{3}\right)$
(D) $75\cos\left(15t + \frac{2\pi}{3}\right)$

Open solution

Q105. NEET 2013 (Karnataka) · MCQ (Single Correct Answer)

One way in which the operation of a n-p-n transistor differs from that of a p-n-p

(A) The emitter junction injects minority carriers into the base region of the p-n-p
(B) The emitter injects holes into the base of the p-n-p and electrons into the base region of n-p-n
(C) The emitter injects holes into the base of n-p-n
(D) The emitter junction is reverse biased in n-p-n

Open solution

Q106. NEET 2013 · MCQ (Single Correct Answer)

In a common emitter (CE) amplifier having a voltage gain G, the transistor used has transconductance 0.03 mho and current gain 25. If the above transistor is replaced with another one with transconductance 0.02 mho and current gain 20, the voltage gain will be

(A) $\frac{1}{3}G$ (B) $\frac{5}{4}G$ (C) $\frac{2}{3}G$ (D) 1.5G

Open solution

Q107. AIPMT 2012 Mains · MCQ (Single Correct Answer)

The input resistance of a silicon transistor is 100 Ω. Base current is changed by 40 μA which results in a change in collector current by 2 mA. This transistor is used as a common emitter amplifier with a load resistance of 4 kΩ. The voltage gain of the amplifier is

(A) 2000 (B) 3000 (C) 4000 (D) 1000

Open solution

Q108. AIPMT 2012 Prelims · MCQ (Single Correct Answer)

In a CE transistor amplifier, the audio signal voltage across the collector resistance of 2 kΩ is 2 V. If the base resistance is 1 kΩ and the current amplification of the transistor is 100, the input signal voltage is

(A) 0.1 V (B) 1.0 V (C) 1 mV (D) 10 mV

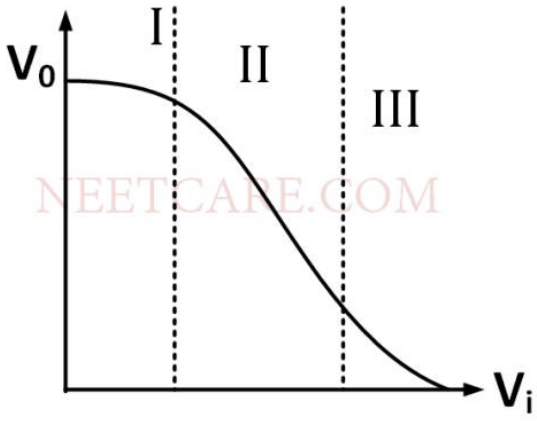
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Q109. AIPMT 2012 Prelims · MCQ (Single Correct Answer)

Transfer characteristics [output voltage (V_0) vs input voltage (V_i)] for a base biased transistor in CE configuration is as shown in the figure. For using transistor as a switch, it is used

Open solution



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(A) in region III
(B) both in region (I) and (III)
(C) in region II
(D) in region I

Q110. AIPMT 2011 Prelims · MCQ (Single Correct Answer)

A transistor is operated in common emitter configuration at V_C = 2 V such that a change in the base current from 100 μA to 300 μA produces a change in the collector current from 10 mA to 20 mA. The current gain is

Open solution

(A) 50
(B) 75
(C) 100
(D) 25

Q111. AIPMT 2010 Mains · MCQ (Single Correct Answer)

For transistor action

(1) Base, emitter and collector regions should have similar size and doping concentrations.
(2) The base region must be very thin and lightly doped.
(3) The emitter-base junction is forward biased and base-collector junction is reverse biased.
(4) Both the emitter-base junction as well as the base-collector junction are forward biased.

Which one of the following pairs of statements is correct?

Open solution

(A) (4) and (1)
(B) (1) and (2)
(C) (2) and (3)
(D) (3) and (4)

Q112. AIPMT 2010 Prelims · MCQ (Single Correct Answer)

A common emitter amplifier has a voltage gain of 50, an input impedance of 100 Ω and an output impedance of 200 Ω. The power gain of the amplifier is

Open solution

(A) 500
(B) 1000
(C) 1250
(D) 50

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Q113. AIPMT 2009 · MCQ (Single Correct Answer)

A transistor is operated in common-emitter configuration at V_C = 2V such that a change in the base current from 100 μA to 200 μA produces a change in the collector current from 5 mA to 10 mA. The current gain is

(A) 100 (B) 150 (C) 50 (D) 75

Open solution

Q114. AIPMT 2008 · MCQ (Single Correct Answer)

The voltage gain of an amplifier with 9% negative feedback is 10. The voltage gain without feedback will be

(A) 1.25 (B) 100 (C) 90 (D) 10

Open solution

Q115. AIPMT 2006 · MCQ (Single Correct Answer)

A transistor is operated in common emitter configuration at constant collector voltage V_C = 1.5 V such that a change in the base current from 100 μA to 150 μA produces a change in the collector current from 5 mA to 10 mA. The current gain β is

(A) 50 (B) 67 (C) 75 (D) 100

Open solution

Q116. AIPMT 2003 · MCQ (Single Correct Answer)

A n-p-n transistor conducts when

(A) both collector and emitter are positive with respect to the base

(B) collector is positive and emitter is negative with respect to the base

(C) collector is positive and emitter is at same potential as the base

(D) both collector and emitter are negative with respect to the base

Open solution

Q117. AIPMT 2002 · MCQ (Single Correct Answer)

For a transistor $\frac{I_C}{I_E}$ = 0.96, then current gain for common emitter is

(A) 12 (B) 6 (C) 48 (D) 24

Open solution

Q118. AIPMT 2001 · MCQ (Single Correct Answer)

For a common base circuit if $\frac{I_C}{I_E}$ = 0.98 then current gain for common emitter circuit will be

(A) 49 (B) 98 (C) 4.9 (D) 25.5.

Open solution

Q119. AIPMT 2000 · MCQ (Single Correct Answer)

The correct relation for α, β for a transistor

(A) $\beta = \frac{1-\alpha}{\alpha}$ (B) $\beta = \frac{\alpha}{1-\alpha}$

(C) $\alpha = \frac{\beta-1}{\beta}$ (D) $\alpha\beta = 1$

Open solution

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Answer Key

MCQ (Single Correct Answer)

1. **C**	2. **D**	3. **D**	4. **D**	5. **B**
6. **D**	7. **B**	8. **D**	9. **B**	10. **C**
11. **C**	12. **A**	13. **D**	14. **B**	15. **D**
16. **D**	17. **D**	18. **B**	19. **A**	20. **B**
21. **D**	22. **D**	23. **C**	24. **C**	25. **C**
26. **D**	27. **D**	28. **C**	29. **A**	30. **D**
31. **B**	32. **A**	33. **B**	34. **D**	35. **B**
36. **D**	37. **A**	38. **C**	39. **B**	40. **D**
41. **C**	42. **A**	43. **B**	44. **A**	45. **D**
46. **A**	47. **B**	48. **A**	49. **A**	50. **B**
51. **D**	52. **C**	53. **A**	54. **A**	55. **C**
56. **A**	57. **D**	58. **D**	59. **C**	60. **D**
61. **B**	62. **B**	63. **A**	64. **B**	65. **C**
66. **B**	67. **A**	68. **D**	69. **B**	70. **C**
71. **C**	72. **C**	73. **D**	74. **B**	75. **D**
76. **C**	77. **C**	78. **D**	79. **A**	80. **B**
81. **B**	82. **A**	83. **B**	84. **B**	85. **C**
86. **B**	87. **D**	88. **A**	89. **C**	90. **B**
91. **A**	92. **A**	93. **A**	94. **B**	95. **B**
96. **A**				

Out of syllabus

97. **D**	98. **D**	99. **C**	100. **D**	101. **B**
102. **B**	103. **C**	104. **B**	105. **B**	106. **C**
107. **A**	108. **D**	109. **B**	110. **A**	111. **C**
112. **C**	113. **C**	114. **B**	115. **D**	116. **B**
117. **D**	118. **A**	119. **B**		

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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